CLASSROOM ATTENDANCE PREDICTION

Using Academic Schedule and Historical Attendance Data

FINAL PROJECT REPORT

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Kaggle Link: https://www.kaggle.com/datasets/pranitaaaaaa/mca-classroom-attendance-datasetraw/

Data Science & Machine Learning Project

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1. Executive Summary

This project presents an end-to-end machine-learning system for predicting classroom attendance percentage from historical attendance and academic scheduling information. The project uses 300 lecture-level records and treats Attendance % as the prediction target. The workflow includes data inspection, cleaning, feature engineering, exploratory data analysis, regression-model training, evaluation, model selection, model saving, and Streamlit deployment.

Four regression algorithms were compared: Linear Regression, Decision Tree Regressor, Random Forest Regressor and Gradient Boosting Regressor. Random Forest was selected as the final model because it recorded the strongest overall evaluation results among the tested models.

Project Objective
Use historical attendance and academic schedule information to estimate expected attendance for an upcoming lecture and support academic planning.

2. Data Collection

The project dataset contains 300 attendance observations. Each record represents a lecture/session and includes academic, scheduling, classroom, faculty and attendance-related information.

The main variables used in the project include:

Feature	Purpose
Date	Exact lecture date
Day of Week	Operational day
Lecture Number	Sequential timetable slot
Start Time	Lecture commencement time
Subject	Academic subject/session
Faculty ID	Encoded instructor identifier
Semester	Current semester
Branch / Section	Academic grouping
Classroom	Physical room identifier
Total Enrolled Students	Class strength
Students Present	Observed attendance count
Attendance %	Prediction target
Previous Lecture Attendance	Historical attendance signal
Practical/Theory	Session format
Test Week / Assignment Due / Holiday	Academic context
Weather / Special Event	Optional context

3. Data Cleaning & Feature Engineering

Data Cleaning

The cleaning stage prepared the raw records for analysis and modeling. Date values were converted into a datetime format, duplicate records were checked, attendance values were validated, and the relationship between Students Present and Total Enrolled Students was checked. Missing historical attendance was handled while retaining a missing-value indicator.

- Duplicate records checked before modeling.
- Attendance values checked for valid percentage range.
- Students Present checked against Total Enrolled Students.
- Missing Previous Lecture Attendance handled using a numeric imputation strategy.
- Faculty experience information was converted into a usable numeric feature.

Feature Engineering

The project derived additional variables to provide temporal and historical signals to the model:

- Day of Semester
- Week Number
- Start Hour
- Time Period
- Rolling 3 Lecture Attendance
- Previous Attendance Missing
- Faculty Experience Years
- Binary Yes/No indicators
- Attendance Category

Why Feature Engineering?

Raw fields such as a date or start time are useful, but derived features can express patterns such as semester progression, morning/afternoon timing and recent attendance behavior in a form that a machine-learning model can use.

4. Exploratory Data Analysis — Key Findings

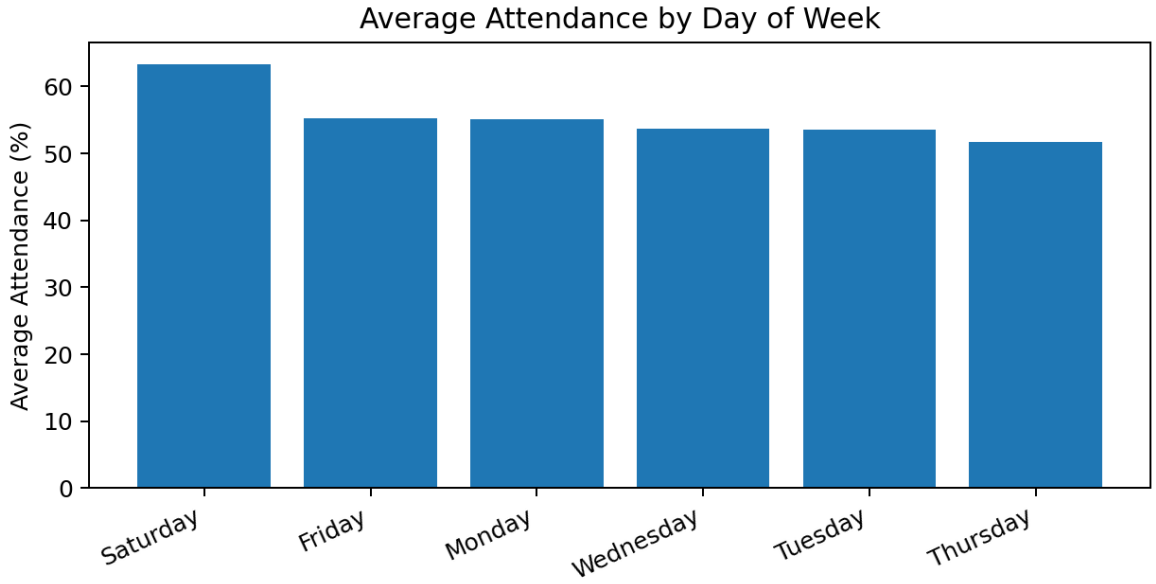
4.1 Overall Distribution

Attendance in the cleaned dataset ranges from 6% to 95%. The mean attendance is 55.43% and the median is 48.00%. This provides the overall baseline against which scheduling and subject-level patterns can be compared.

Key Finding
Overall average attendance = 55.43%.

4.2 Day of Week

Saturday has the highest average attendance (63.37%) in the supplied sample, while Thursday has the lowest (51.70%). These values describe this dataset and should not be interpreted as universal causal effects.



4.3 Subject-wise Attendance

Subject-level averages vary across the dataset. Talenlio ERP Hands-on has the highest observed subject average (80.92%), while Industry Readiness Program has the lowest (35.38%). Subject-level analysis is useful for identifying sessions that may require additional engagement or scheduling attention.

Subject	Average Attendance	Observation
Talenlio ERP Hands-on	80.92%	Higher observed average
Regular Afternoon Session (Delayed)	78.17%	Higher observed average
Data Science & Machine Learning Hands-on	75.62%	Higher observed average
Mobile Application Development Hands-on	70.60%	Higher observed average

Project Milestone 1 Evaluation	68.67%	Higher observed average
Industry Readiness Program	35.38%	Lower observed average
Mobile Application Development	41.89%	Lower observed average
Data Science & Machine Learning	47.48%	Lower observed average
Software Testing & Quality Assurance	48.25%	Lower observed average
Principles of Cloud Management and Security	49.03%	Lower observed average

4.4 Effect of Test Weeks, Weather, and Holidays

Weather averages in the available records are: Cloudy: 56.02%; Rainy: 55.01%. The dataset contains 0 records marked as Test Week=Yes and 0 records marked as Holiday=Yes. Because there are no positive Test Week or Holiday observations, their effects cannot be reliably estimated from this dataset. This limitation is important when interpreting the EDA.

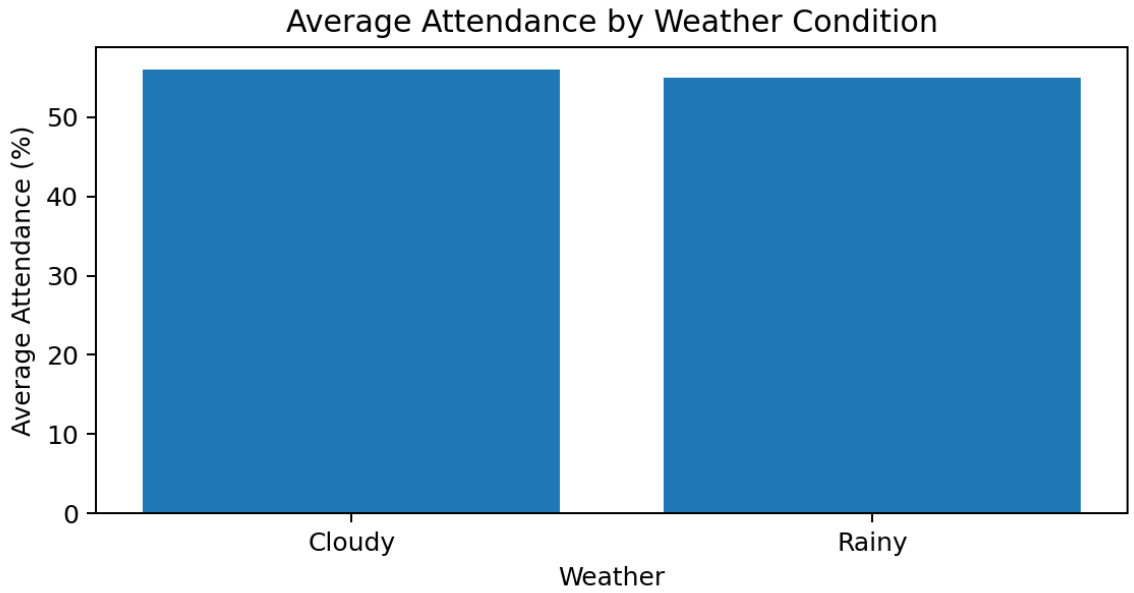


Figure 4.4a. Average attendance by weather condition.

4.5 Trend Over Time

Attendance varies across the observation period rather than remaining constant. The project therefore uses temporal features and recent attendance history as predictive signals. The Rolling 3 Lecture Attendance feature summarizes recent attendance behavior without directly using the current target value.

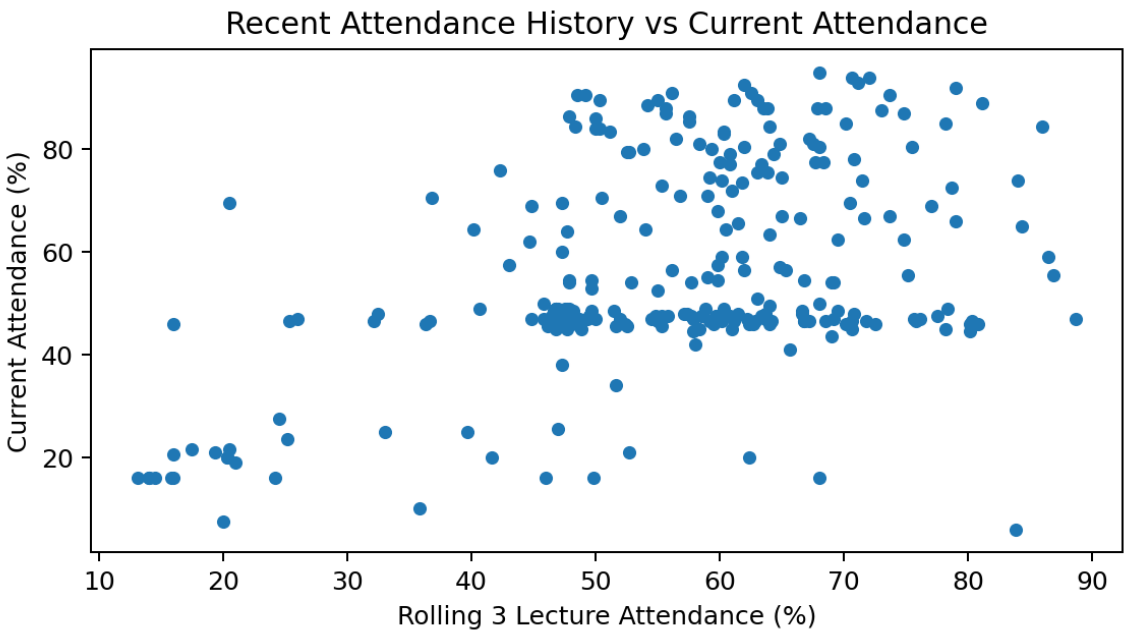


Figure 4.5b. Recent attendance history versus current attendance.

5. Modeling Approach & Results + Diagrams

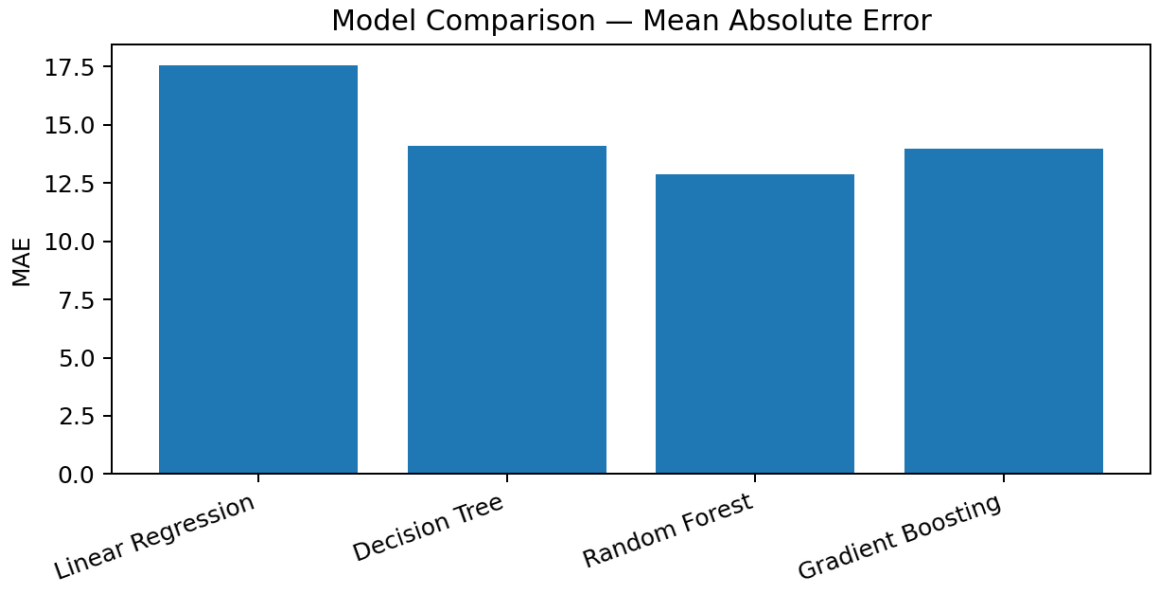
The implemented task is regression because Attendance % is a continuous numerical target. The dataset was split into training and testing portions using an 80:20 ratio. Categorical inputs were transformed using One-Hot Encoding, while numerical preprocessing was incorporated into the Scikit-learn pipeline.

Modeling Pipeline

Data → Train/Test Split → Preprocessing → Regression Models → MAE/RMSE/MAPE/R^2 → Model Comparison → Random Forest → Saved Model → Streamlit App

5.1 Regression Results

Model	MAE	RMSE	MAPE	R^2
Linear Regression	17.57	22.37	29.28%	-0.9702
Decision Tree	14.10	15.62	23.08%	0.0386
Random Forest	12.87	14.80	20.92%	0.1369
Gradient Boosting	13.96	16.90	23.06%	-0.1253



Selected Model — Random Forest Regressor

MAE = 12.87 | RMSE = 14.80 | MAPE = 20.92% | R^2 = 0.1369

Random Forest achieved the lowest MAE, RMSE and MAPE and the highest R^2 among the four tested regression models. Therefore it was selected as the best-performing model for deployment. The R^2 value is modest, which means that substantial variation in attendance remains unexplained; the system should be treated as a decision-support prototype rather than a perfect forecast.

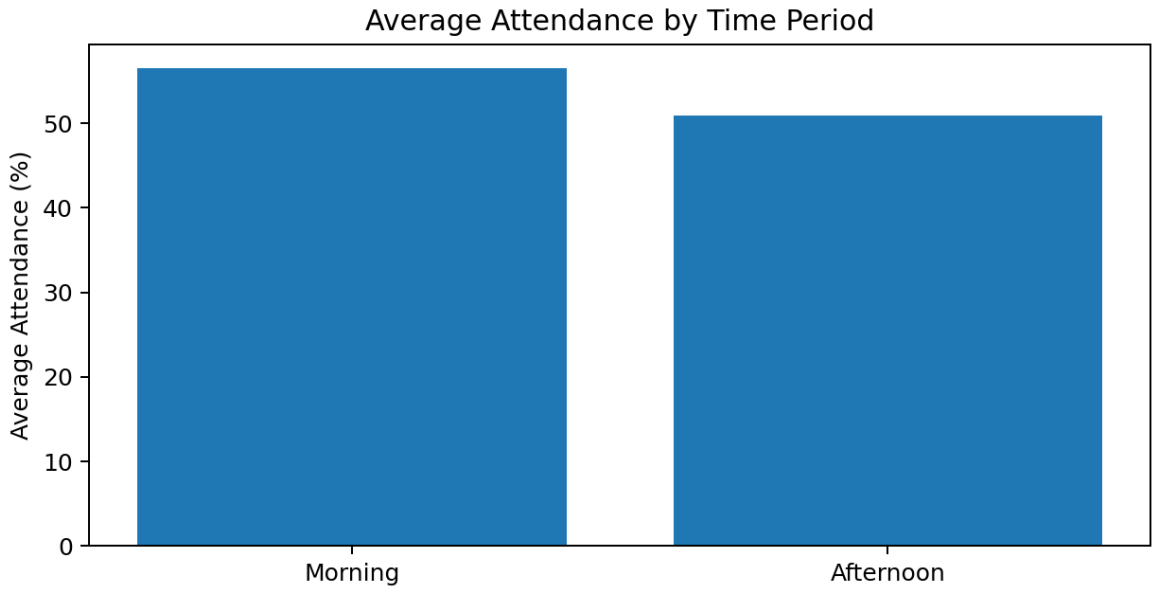
5.2 Classification Results

The project also defines Attendance Category bands — Low (<50%), Medium (50–75%) and High (>75%) — during feature engineering. However, the submitted machine-learning implementation does not train a separate classification model or report classification metrics. Therefore, Accuracy, Precision, Recall, F1 Score and ROC-AUC results are not reported. The deployed predictive system is regression-based.

6. Optimal Scheduling Recommendations

The following recommendations are based on observed associations in the supplied dataset:

- Prefer morning slots when higher attendance is desired: Morning average = 56.45%, Afternoon average = 50.95%.
- Saturday recorded the highest day-level average attendance (63.37%) in this sample.
- Thursday recorded the lowest day-level average attendance (51.70%) in this sample.
- Practical sessions showed higher average attendance (61.38%) than theory sessions (49.99%).
- Subjects with lower historical attendance should receive additional engagement or scheduling attention.
- The prediction application can be used to compare expected attendance for alternative lecture schedules.



7. Deployment Strategy

The final project includes a Streamlit application that uses the saved machine-learning model to provide interactive attendance predictions. The application loads the trained model and the feature-information metadata, accepts lecture and academic inputs, applies the same feature structure used during training, and displays the predicted attendance percentage.

Component	Role
app/app.py	Streamlit user interface and prediction logic
models/best_model.pkl	Saved trained model

models/feature_information.json	Expected feature information/metadata
data/cleaned/attendance_engineered.csv	Engineered modeling dataset
outputs/results/model_comparison.csv	Experimental model comparison results

User Flow
Enter lecture details → Prepare features → Load saved model → Generate prediction → Display expected attendance percentage.

8. Conclusion

This project demonstrates a complete classroom attendance prediction workflow, from data preparation and exploratory analysis to machine-learning model comparison and interactive deployment. Random Forest was the strongest of the four tested regression models according to the recorded evaluation results.

The current dataset has limitations, particularly its relatively small size and the absence of positive Test Week and Holiday observations. Attendance is also influenced by factors that may not be fully captured by the available variables. A larger dataset collected across more classes, divisions, semesters and academic conditions would improve the reliability and generalizability of future versions.